

# STAT 220B Cheatsheet: The Story of the Course

## Decision Theory so far

This cheatsheet traces the **logical flow** of the course. For each concept, three questions: **why** is it needed, **how** does it connect to what came before, and **when** do you reach for it. It is a map, not a substitute for the lectures.

## The five ingredients of every decision problem

Every problem from Lecture 1 onward has the same five components.

| Symbol                               | Name             | What it is                                                             |
|--------------------------------------|------------------|------------------------------------------------------------------------|
| $\Theta$                             | Parameter space  | Set of possible truths $\theta$                                        |
| $\mathcal{A}$                        | Action space     | Set of possible decisions                                              |
| $L(\theta, a)$                       | Loss function    | Cost of action $a$ when truth is $\theta$ . Bounded below: $L \geq -K$ |
| $X \in \mathcal{X}, X \sim P_\theta$ | Data and model   | Observations whose distribution depends on $\theta$                    |
| $\pi(\theta)$                        | Prior (optional) | Distribution on $\Theta$ encoding belief or weighting                  |

A **decision rule**  $\delta : \mathcal{X} \rightarrow \mathcal{A}$  takes data to actions. A **randomized rule**  $\delta^*(x, \cdot)$  is a distribution on  $\mathcal{A}$  for each  $x$ . In no-data problems, a rule is just an action.

## Two viewpoints for evaluating a rule

The most important table in the course. Every notation confusion traces back to forgetting which side you are on.

|                  | Frequentist viewpoint                                                                                                                    | Bayesian viewpoint                                                             |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| Average over     | Data $X$ , holding $\theta$ <i>fixed</i>                                                                                                 | Parameter $\theta$ , possibly given $X$                                        |
| Main object      | $R(\theta, \delta) = \mathbb{E}_{P_\theta}[L(\theta, \delta(X))]$<br><i>risk function</i>                                                | $\rho(\pi, a) = \mathbb{E}^\pi[L(\theta, a)]$<br><i>Bayesian expected loss</i> |
| Returns          | A <i>function</i> of $\theta$                                                                                                            | A <i>number</i> for each action                                                |
| Compare rules by | Partial order: $\delta_1$ R-better than $\delta_2$ iff $R(\theta, \delta_1) \leq R(\theta, \delta_2)$ for all $\theta$ , strict for some | Direct comparison: pick the smallest                                           |
| Bridge via prior | $r(\pi, \delta) = \mathbb{E}^\pi[R(\theta, \delta)]$ , the <i>Bayes risk</i> of $\delta$                                                 | Average posterior expected loss over the marginal $m(x)$ of $X$                |

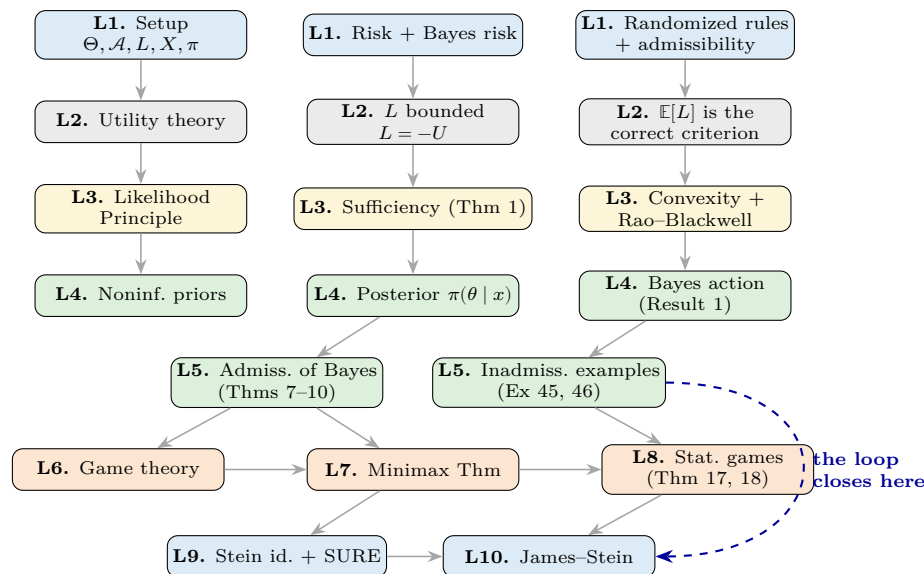
**Notation tag (the thing that trips everyone).**  $\mathbb{E}_{P_\theta}[\cdot]$  averages over the data with  $\theta$  fixed.  $\mathbb{E}^\pi[\cdot]$  averages over  $\theta$  with prior or posterior  $\pi$ .  $\mathbb{E}[\cdot \mid T = t]$  is conditional over data given a sufficient statistic. **Test:** ask “what is random in this expression, and what is fixed?”

**The crucial identity (Result 1 of §4.4, used in every Bayes calculation):**

$$r(\pi, \delta) = \mathbb{E}^\pi[R(\theta, \delta)] = \mathbb{E}^{m(x)}[\rho(\pi(\theta \mid x), \delta(x))].$$

A Bayes rule can be found by minimizing posterior expected loss pointwise in  $x$  (“extensive form”). This is what you use in practice.

## The course as a flow diagram



**Color key.** Blue framework and Stein. Gray foundations (utility). Yellow structural tools. Green Bayesian / admissibility. Orange minimax. The **dashed arrow** is the punchline: Example 46 in Lecture 5 flagged that the sample mean is inadmissible for  $p \geq 3$ . Lectures 9–10 close the loop by proving it.

## Narrative: why each layer exists

**Layer 1: Setup (Lecture 1).** The five ingredients, the risk function  $R(\theta, \delta)$  and Bayes risk  $r(\pi, \delta)$ , admissibility, randomization, and three selection principles (Bayes, minimax, invariance). *Why needed:* Statistics is not just estimation. It is deciding under uncertainty, with consequences. We need a framework that handles the cost of being wrong, not just the standard error. *Unsolved problem at end:* risk functions cross, so admissibility alone does not pick a unique rule. We need selection principles, plus tools to simplify the search.

**Layer 2: Utility theory, the foundation under everything (Lecture 2).** *Why it exists (and why it did not feel obvious in lecture).* Lecture 1 takes  $L(\theta, a)$  as a given input and announces that we will minimize  $\mathbb{E}[L]$ . Two questions are left dangling: (1) Where does  $L$  come from? (2) Why is minimizing  $\mathbb{E}[L]$  the right thing to do, rather than, say, the median loss, the worst-case loss, or some other functional? Utility theory answers both, and provides a property of  $L$  that the next four layers all silently use.

*The motivating puzzle.* Offered (A) \$200,000 for certain, or (B) a 50–50 chance of \$0 or \$410,000, most people pick (A) even though (B) has higher expected dollars. “Value” is not linear in money. We need a numerical encoding of preferences that captures this. That is what a utility function  $U$  does.

*The axioms (von Neumann–Morgenstern).* Preferences over distributions of consequences satisfy: (1) completeness, (2) transitivity, (3) independence (mixing with a third distribution preserves preference), (4) continuity (no consequence is infinitely good or bad). Under these axioms:

1. There exists a utility function  $U : \mathcal{R} \rightarrow \mathbb{R}$  such that  $P_1 \prec P_2 \iff \mathbb{E}^{P_1}[U] < \mathbb{E}^{P_2}[U]$ . Unique up to positive affine transformation.
2.  $U$  is **bounded** (forced by Axiom 4; the St. Petersburg paradox is the demonstration).
3. Define  $L(\theta, a) = -U(\theta, a)$ . Then maximizing  $\mathbb{E}[U]$  is the same as minimizing  $\mathbb{E}[L]$ .

*Why utility matters more than it looks.* Three downstream consequences, each invoked later without explicit reference back here.

- It **justifies expected loss** as the criterion. Minimizing  $\mathbb{E}[L]$  is not one option among many. It follows from the axioms.
- It **forces  $L$  to be bounded below by  $-K$** , which is exactly the assumption that makes the admissibility theorems of Lecture 5 work (Theorems 7–9 all need finite Bayes risk, which needs  $L$  bounded).

- It **resolves the St. Petersburg paradox**: naive expected monetary value gives an infinite price to play, but bounded utility removes the pathology. The same fix prevents formal Bayes pathologies later.

*A variant worth knowing: regret loss.*  $\ell(\theta, a) = \sup_{a'} U(\theta, a') - U(\theta, a)$ . Penalty relative to the best possible action. For Bayesian analysis, equivalent to  $L = -U$  (they differ by a function of  $\theta$  alone, which does not affect minimization). Can differ for minimax, where we return to it in Chapter 5.

*Short version, in one sentence.* Utility is the load-bearing wall behind the framework. The rest of the course assumes its conclusions ( $\mathbb{E}[L]$  is the criterion,  $L$  is bounded), uses them, and never looks back.

**Layer 3: Simplifying tools (Lecture 3).** The decision rule space  $\mathcal{D}^*$  is enormous. Three theorems shrink the search before any optimization begins.

- **Likelihood Principle (motivating idea).** The 9-heads / 3-tails binomial vs. negative-binomial example: same likelihood, different p-values. Classical analysis can violate the LP. The LP suggests all relevant information about  $\theta$  lives in the likelihood, which lives in a sufficient statistic.
- **Sufficiency (Theorem 1).** If  $T$  is sufficient for  $\theta$ , every rule  $\delta_0$  is R-equivalent to some rule based on  $T$ . *Search only over rules of  $T$ .*
- **Convexity + Jensen  $\Rightarrow$  Theorem 3.** If  $L$  is convex in  $a$ , any randomized rule is dominated (in pointwise loss) by its mean. *Search only over nonrandomized rules.*
- **Rao–Blackwell (Theorem 4).** Combines both:  $\delta_1(t) = \mathbb{E}[\delta_0(X) \mid T = t]$  is nonrandomized, based on  $T$ , and R-better than  $\delta_0$  whenever the loss is convex.

*Connection upward.* Theorem 3 needs  $L$  convex in  $a$ . Where did convexity become a reasonable assumption? Utility theory (Layer 2) put  $L$  in the picture as a quantitative loss; convex losses (squared error, absolute error, log loss) are then natural choices. *Connection downward:* every Bayes, minimax, and Stein rule we build below quietly uses this reduction. You never write down a randomized rule again unless an adversary is involved.

**Layer 4: Bayesian decision theory (Lecture 4).** The mechanics of computing optimal rules given a prior.

- **Noninformative priors** for when no prior information is available: location  $\pi(\theta) = 1$ , scale  $\pi(\sigma) = 1/\sigma$ , Bernoulli Jeffreys  $\pi(p) \propto [p(1-p)]^{-1/2}$ . All improper.
- **Posterior**  $\pi(\theta \mid x) \propto f(x \mid \theta)\pi(\theta)$ . For improper priors this is *formal*, valid if the marginal  $m(x)$  is finite.
- **Posterior expected loss**  $\rho(\pi(\theta \mid x), a) = \mathbb{E}^{\pi(\theta \mid x)}[L(\theta, a)]$  and the **Bayes action**  $\delta^\pi(x) = \arg \min_a \rho(\pi(\theta \mid x), a)$ .
- **Result 1** (extensive form = normal form): a rule is Bayes iff it minimizes posterior expected loss pointwise in  $x$ . This is the identity from Section 2 above, and it is the single most-used tool in the course.

Three workhorse Bayes actions, one for each common loss:

| Loss                             | Bayes action                               | Why                                                                                 |
|----------------------------------|--------------------------------------------|-------------------------------------------------------------------------------------|
| $(\theta - a)^2$ (squared error) | Posterior mean $\mathbb{E}[\theta \mid x]$ | Variance decomposition: $\arg \min \mathbb{E}[(\theta - a)^2] = \mathbb{E}[\theta]$ |
| $ \theta - a $ (absolute error)  | Posterior median                           | Calculus on $\int  t - a  dF(t)$                                                    |
| 0–1 loss                         | Posterior mode (MAP)                       | Direct minimization on indicator loss                                               |

**Layer 5: Admissibility of Bayes rules (Lecture 5).** The question: are Bayes rules admissible?

*The good news* (§4.8.1, proper priors with finite Bayes risk). Theorem 7: discrete  $\Theta$ , prior charges every point  $\Rightarrow$  admissible. **Theorem 8: unique Bayes rule  $\Rightarrow$  admissible** (used most often). Theorem 9: continuous risks, prior charges every open set  $\Rightarrow$  admissible. *All three need finite Bayes risk, which is where bounded loss from Layer 2 (utility theory) shows up.*

*The bad news* (§4.8.2, improper priors / generalized Bayes rules). When  $r(\pi, \delta) = \infty$ , the Bayes-risk argument fails. Two cautionary examples:

- **Example 45 (gamma scale).**  $\pi(\beta) = 1/\beta$  for Gamma( $\alpha, \beta$ ). The generalized Bayes rule is dominated by a constant multiple, with risk ratio  $(\alpha + 1)^2/(\alpha - 1)^2$ .
- **Example 46 (Stein).**  $\pi(\theta) = 1$  on  $\mathbb{R}^p$  with  $\mathbf{X} \sim \mathcal{N}_p(\theta, I_p)$ . The generalized Bayes rule  $\delta^0(\mathbf{x}) = \mathbf{x}$  is inadmissible for  $p \geq 3$ . *This is the example Lectures 9–10 will actually prove.*

*Long-run motivation* (§4.8.3). Even a Bayesian should care about admissibility: using an inadmissible rule repeatedly produces, with probability one, a higher long-run average loss than a dominating rule. Admissibility is not a Bayesian-vs-frequentist issue; it is a sanity check.

**Layer 6: Minimax theory (Lectures 6–8).** If no prior is available, choose  $\delta$  to minimize  $\sup_{\theta} R(\theta, \delta)$ , protecting against the worst case.

- **L6 (game theory).** Two-person zero-sum games, randomized strategies, upper and lower values. Theorem 2 (verification by inequality); Theorem 4 (carrier of optimal strategies).
- **L7 (Minimax Theorem, Thm 15).** For finite  $\Theta$  and bounded loss, the game has a value, and maximin and minimax strategies exist. Proved via S-game reformulation and separating hyperplanes (supplementary notes).
- **L8 (statistical games).** Theorem 17 (equalizer + Bayes  $\Rightarrow$  minimax) and Theorem 18 (sequence of priors for improper limits).

*Practical recipe.* (1) Try a conjugate family  $\pi_{\alpha}$ . (2) Compute Bayes rule  $\delta^{\pi_{\alpha}}$ . (3) Compute risk  $R(\theta, \delta^{\pi_{\alpha}})$ . (4) Solve for  $\alpha$  that makes  $R$  constant in  $\theta$ . (5) Conclude (Thm 17):  $\delta^{\pi_{\alpha}}$  is minimax,  $\pi_{\alpha}$  is least favorable.

Two foundational examples to keep in mind:

| Problem                                                       | Minimax rule                                  | Prior used                                                              |
|---------------------------------------------------------------|-----------------------------------------------|-------------------------------------------------------------------------|
| $X \sim \text{Bin}(n, p)$ , squared loss                      | $\delta(x) = (x + \sqrt{n}/2)/(n + \sqrt{n})$ | Beta( $\sqrt{n}/2, \sqrt{n}/2$ ) (proper; Thm 17)                       |
| $\bar{X} \sim \mathcal{N}(\theta, \sigma^2/n)$ , squared loss | $\delta(\bar{x}) = \bar{x}$ (sample mean)     | $\mathcal{N}(0, \tau^2)$ , $\tau \rightarrow \infty$ (improper; Thm 18) |

**Layer 7: The Stein phenomenon (Lectures 9–10).** The setup left over from Lecture 5.  $\mathbf{X} \sim \mathcal{N}_p(\theta, I_p)$ , squared-error loss. The sample mean  $\delta^0(\mathbf{x}) = \mathbf{x}$  is MLE, UMVUE, minimax, with constant risk  $p$ . But Example 46 said it is inadmissible for  $p \geq 3$ . How is that possible?

- **Stein’s identity (L9).** For  $\mathbf{X} \sim \mathcal{N}_p(\theta, I_p)$  and smooth  $\mathbf{g}$ :  $\mathbb{E}_{\theta}[(\mathbf{X} - \theta)^T \mathbf{g}(\mathbf{X})] = \mathbb{E}_{\theta}[\nabla \cdot \mathbf{g}(\mathbf{X})]$ . Proved by integration by parts. Eliminates  $\theta$  from risk computations.
- **SURE (L9).** For  $\delta(\mathbf{x}) = \mathbf{x} + \mathbf{g}(\mathbf{x})$ :  $\hat{R}(\mathbf{x}) = p + 2\nabla \cdot \mathbf{g}(\mathbf{x}) + \|\mathbf{g}(\mathbf{x})\|^2$  is an unbiased estimator of risk. Risk dominance reduces to a pointwise comparison of SURE.
- **James–Stein (L10).** With  $\mathbf{g}(\mathbf{x}) = -(p - 2)\mathbf{x}/\|\mathbf{x}\|^2$ :  $\hat{R}^{JS}(\mathbf{x}) = p - (p - 2)^2/\|\mathbf{x}\|^2$ , hence  $R(\theta, \delta^{JS}) = p - (p - 2)^2\mathbb{E}_{\theta}[1/\|\mathbf{X}\|^2] < p$  for all  $\theta$ .

*Why it matters beyond the proof.* Stein’s phenomenon is the foundation of modern shrinkage: empirical Bayes, ridge regression, lasso, hierarchical models, wavelet denoising (SureShrink). In each case, high-dimensional geometry creates statistical leverage that one-dimensional intuition does not see.

## Cross-reference: which tool to reach for

| If you are asked to...                                  | Use...                                                                                                                                              |
|---------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Compute risk of a specific rule $\delta$                | Definition: $R(\theta, \delta) = \mathbb{E}_{P_\theta}[L(\theta, \delta(X))]$ . For squared loss, bias–variance: $R = \text{Bias}^2 + \text{Var}$ . |
| Find the Bayes rule for given $\pi$                     | <b>Extensive form</b> (Result 1, §4.4): minimize posterior expected loss for each $x$ . For squared loss, the posterior mean.                       |
| Show a rule is admissible                               | Thm 8 (unique Bayes rule): show it minimizes Bayes risk uniquely. Or Thms 7 / 9 for the technical conditions.                                       |
| Show a rule is inadmissible                             | Construct a dominating rule. For Stein-type problems, compare SURE pointwise.                                                                       |
| Improve a non- $T$ -based estimator                     | Rao–Blackwell (Thm 4): condition on the sufficient statistic $T$ .                                                                                  |
| Find a minimax rule (proper prior)                      | Thm 17: find $\pi$ such that the Bayes rule $\delta^\pi$ has <i>constant risk</i> in $\theta$ .                                                     |
| Find a minimax rule (improper prior)                    | Thm 18: find proper priors $\pi_n$ with $\lim r(\pi_n) = c < \infty$ , and a rule $\delta^*$ with $R(\theta, \delta^*) \leq c$ everywhere.          |
| Prove a Stein-type dominance                            | Stein’s identity + SURE: compute $\hat{R}(\mathbf{x}) = p + 2\nabla \cdot \mathbf{g} + \ \mathbf{g}\ ^2$ and compare pointwise.                     |
| Decide whether minimizing $\mathbb{E}[L]$ is reasonable | Utility theory (Layer 2): it is forced by the rationality axioms.                                                                                   |

## Five common confusions

1. **“Which expectation?”**  $\mathbb{E}_{P_\theta}$  vs.  $\mathbb{E}^\pi$  vs.  $\mathbb{E}^{\pi(\theta|x)}$ . Ask: is  $\theta$  fixed or random? Frequentist risk fixes  $\theta$ . Posterior expected loss fixes  $x$  and treats  $\theta$  as random. Bayes risk averages over both.
2. **“Improper prior  $\Rightarrow$  no Bayes rule?”** No: you get a *generalized Bayes rule*. The posterior  $\pi(\theta | x) \propto f(x | \theta)\pi(\theta)$  is computed the same way. What fails is the Bayes-risk machinery ( $r(\pi, \delta) = \infty$ ); admissibility (Thms 7–9) and minimaxity (Thm 17) need Thm 18 as a workaround.
3. **“Why is utility in this course?”** Three uses: (a) it *justifies*  $\mathbb{E}[L]$  as the right criterion; (b) it *forces*  $L$  bounded, which makes Thms 7–9 work; (c) it *provides the formal recipe*  $L = -U$ . Without it, the framework is ad hoc. See Layer 2 above.
4. **Sufficiency vs. Bayes vs. minimax.** Sufficiency is a property of the *data* (a statistic on  $\mathcal{X}$ ). Bayes and minimax are *selection principles for rules*. Sufficiency reduces the search space (every rule has a replacement based on  $T$ ); Bayes and minimax pick a specific rule within the reduced space.
5. **Bayes rule vs. Bayes action.** A *Bayes action* is for a fixed  $x$ : the action minimizing posterior expected loss. A *Bayes rule* is the function  $\delta^\pi(x)$  returning the Bayes action for each  $x$ . Result 1 says these are equivalent: pointwise minimization gives a rule that also minimizes the Bayes risk.

## Equation reference card

| Object                          | Formula                                                                                                                                            |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Frequentist risk                | $R(\theta, \delta) = \mathbb{E}_{P_\theta}[L(\theta, \delta(X))]$                                                                                  |
| Bayes risk of $\delta$          | $r(\pi, \delta) = \mathbb{E}^\pi[R(\theta, \delta)] = \int R(\theta, \delta) dF^\pi(\theta)$                                                       |
| Posterior expected loss         | $\rho(\pi(\theta   x), a) = \mathbb{E}^{\pi(\theta x)}[L(\theta, a)]$                                                                              |
| Posterior                       | $\pi(\theta   x) = f(x   \theta)\pi(\theta)/m(x), \quad m(x) = \int f(x   \theta)\pi(\theta) d\theta$                                              |
| Bayes rule (squared loss)       | $\delta^\pi(x) = \mathbb{E}^{\pi(\theta x)}[\theta]$ (posterior mean)                                                                              |
| Result 1 identity               | $r(\pi, \delta) = \mathbb{E}^{m(x)}[\rho(\pi(\theta   x), \delta(x))]$                                                                             |
| Loss from utility               | $L(\theta, a) = -U(\theta, a)$ (regret: $\ell(\theta, a) = \sup_{a'} U(\theta, a') - U(\theta, a)$ )                                               |
| Rao–Blackwell rule              | $\delta_1(t) = \mathbb{E}[\delta_0(X)   T = t]$                                                                                                    |
| Stein's identity (Gaussian)     | $\mathbb{E}_\theta[(\mathbf{X} - \theta)^T \mathbf{g}(\mathbf{X})] = \mathbb{E}_\theta[\nabla \cdot \mathbf{g}(\mathbf{X})]$                       |
| SURE                            | $\hat{R}(\mathbf{x}) = p + 2\nabla \cdot \mathbf{g}(\mathbf{x}) + \ \mathbf{g}(\mathbf{x})\ ^2$ for $\delta = \mathbf{x} + \mathbf{g}(\mathbf{x})$ |
| James–Stein estimator           | $\delta^{JS}(\mathbf{x}) = (1 - (p-2)/\ \mathbf{x}\ ^2)\mathbf{x}$                                                                                 |
| James–Stein risk ( $p \geq 3$ ) | $R(\theta, \delta^{JS}) = p - (p-2)^2 \mathbb{E}_\theta[1/\ \mathbf{X}\ ^2]$                                                                       |
| Binomial minimax                | $\delta(x) = (x + \sqrt{n}/2)/(n + \sqrt{n})$                                                                                                      |

## What is coming next

- **Invariance (Berger Ch 6).** The third selection principle. If the problem has a symmetry, restrict to equivariant rules. Connects to Stein:  $\mathcal{N}_p(\theta, I_p)$  has translation invariance, the best invariant rule is  $\bar{X}$ , which Lecture 10 showed is inadmissible for  $p \geq 3$ . Invariance and admissibility can conflict, in interesting ways.
- **Sequential analysis (Ch 7).** Optimal stopping. Builds on the Bayes machinery of Layer 4.
- **Complete classes (Ch 8).** Under regularity conditions, admissible rules are essentially Bayes rules. Closes the loop between admissibility (Layer 5) and Bayes machinery (Layer 4).

*If something here does not click, come to office hours or send me a question.*  
*– Arka*